

Idemia Datachallenge (IADATA704): Face Occlusion Prediction

Group Name: Team GPU

Group Number: 21

Team Members: Oscar DE LA CRUZ, Rémi HAMON

Project repository: <https://github.com/odelacruz-fierro/data-challenge-idemia.git>

1 Introduction

In the context of the Data Challenge proposed by Idemia, the objective is to predict the percentage of facial occlusion in "in the wild" images (224x224 pixels). The major challenge lies in handling the training data imbalance (a low proportion of highly occluded faces) and the strict gender fairness requirement imposed by the evaluation metric.

In this report, we present our solution based on multi-task learning, designed to extract robust features while minimizing the demographic bias penalty.

2 Exploratory Data Analysis

The exploratory data analysis performed on the train database revealed that only a low proportion of the faces show a high occlusion value ≈ 0.4 (see Fig.1) and an unbalance distribution in terms of gender, i.e. Men = 67.6 % and Women = 32.4 % (see Fig.2).

From Fig.1 we can observe a right-skewed distribution. We can observe a longer tail on the right side, which indicates that the concentration is more towards the lower values and the extreme high values (outliers) drag the tail towards the right side.

In order to better understand the visual impact of the occlusion levels, random samples of different occlusion intervals were inspected (see Fig.3). Even though inspection was not exhaustive, we can note a couple of interesting observations: meantime faces with a Low level of occlusion look normal, faces with Medium level occlusion will be impacted by hair, hands/arms or a hat. Finally the images with a High level of occlusion are not necessarily covered but kind of blurry. If we assume these factors as representative of the whole dataset (train + test) this learning could be useful if doing data augmentation.

3 Methodology & Architecture

TO DO ...

3.1 Milti-Task Learning (MLT)

TO DO ...

3.2 Backbone architectures

TO DO ...

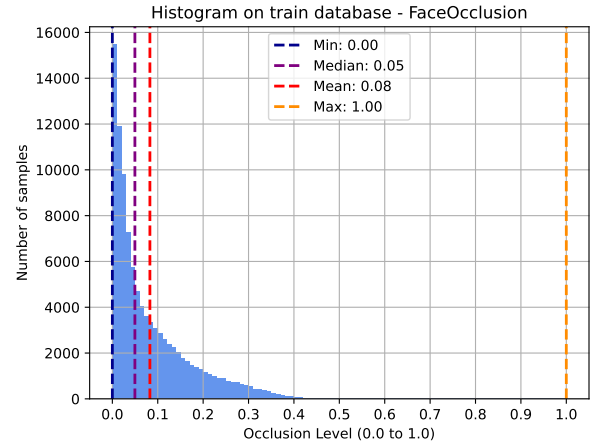


Figure 1: Distribution of occlusion across the training dataset.

4 Training strategy

TO DO ...

4.1 Custom loss

TO DO ...

4.2 Data augmentation

5 Results

TO DO ...

6 Limitations and tradeoffs

7 Conclusion

TO DO ...

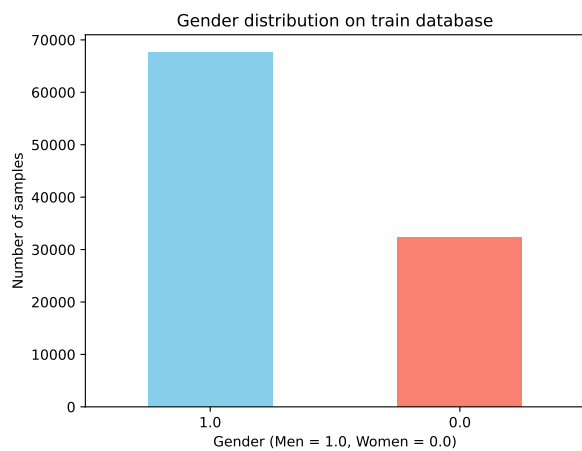


Figure 2: Distribution of gender across the training dataset: Men = 67.6 % and Women = 32.4 %.



Figure 3: Random samples for three occlusion intervals: Low $\in [0.0, 0.3]$, Medium $\in [0.31, 0.6]$ and High $\in [0.61, 1.0]$.